

Session 02 (AIML) - Assignment Questions

Q1. (if Statement)

Write a program that asks the user to enter their age.

If the age is **18 or more**, print the message: "You are eligible to vote."

(Do not print anything if the condition is False).

Add comments.

Q2. (if-else Statement)

Write a program that takes a number as input from the user and checks whether it is **even** or **odd**.

Print "Even Number" or "Odd Number" accordingly.

Q3. (if-elif-else Statement)

Write a program to assign grades based on marks entered by the user:

- Marks ≥ 90 → "Grade A - Excellent"
- Marks ≥ 75 → "Grade B - Good"
- Marks ≥ 60 → "Grade C - Average"
- Marks ≥ 40 → "Grade D - Pass"
- Marks < 40 → "Fail"

Use if-elif-else and print the appropriate grade message.

Q4. (For Loop with range)

Write a program using for loop and range() to:

- a) Print all numbers from 1 to 30.
- b) Print all **odd numbers** from 1 to 50.
- c) Print the sum of numbers from 1 to 15.

Add comments explaining the use of range().

Q5. (While Loop)

Write a program using while loop to print the following:

- Numbers from 1 to 20.
- Also print the **cube** of each number (i.e., 1^3 , 2^3 , 3^3 , ...).

Stop when the number exceeds 20.

Q6. (While Loop - User Controlled)

Write a program that repeatedly asks the user to enter a positive number.

Keep adding the entered numbers to a running total.

Stop when the user enters a negative number or zero.

Finally, print the total sum.

Q7. (Nested if Statement)

Write a program that takes two inputs:

- temperature (integer)
- is_raining (enter "yes" or "no")

Then give advice using nested if:

- If temperature > 30 and not raining → "Hot day, wear light clothes."
 - If temperature > 30 and raining → "Hot and rainy, carry umbrella."
 - If temperature < 15 and raining → "Cold and rainy, wear jacket and take umbrella."
 - Else give a suitable message.
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Q8. (For Loop + if-elif-else)

Write a program using for loop that goes from 1 to 40 and:

- Prints "Fizz" if number is divisible by 3
 - Prints "Buzz" if number is divisible by 5
 - Prints "FizzBuzz" if divisible by both 3 and 5
 - Otherwise prints the number
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Q9. (Menu Driven Program)

Create a simple calculator using while loop and if-elif-else:

Display this menu repeatedly:

1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit

- Ask user for choice.
- Take two numbers as input.
- Perform the operation.
- Continue until user chooses 5 (Exit).
- Handle division by zero case.

Q10. (Debugging)

The following code contains several errors. Correct the program and add meaningful comments explaining each fix:

```
num = input("Enter a number: ")

if num > 100
    print("Large number")
elif num > 50
    print("Medium number")
else
    print("Small number")

count = 1
while count < 10
print(count)
    count + 1
```